

Summary: Introduction to Agentic AI

This document provides an overview of agentic AI, its rationale, evolution, and foundational design patterns. Below is a concise summary of the key points, with references to relevant pages.

1. From Next Token Predictors to Thinking Machines (pp. 3-18)

- Early large language models (LLMs) functioned as "next token predictors," generating text based on statistical patterns (pp. 5-13).
- Limitations included lack of true reasoning, inability to perform calculations, and no verification of outputs (p. 14).
- Advances such as chain-of-thought prompting, tool use, and reasoning architectures have enabled LLMs to solve more complex tasks and improve accuracy (pp. 19-22).

2. Reasoning and Context in LLMs (pp. 23-36)

- Modern LLMs incorporate reasoning steps, planning, and self-reflection, moving beyond linear, statistical thinking (pp. 24-26).
- Context is crucial: LLMs require relevant information and memory (short-term, long-term, parametric) to answer complex queries (pp. 31-34).
- Retrieval-augmented generation (RAG) combines LLMs with external knowledge sources, enhancing their ability to provide accurate, context-aware responses (pp. 35-36).

3. What is Agentic AI? (pp. 38-44)

- Agentic AI systems are characterized by autonomy, goal management, memory, reasoning, planning, and the ability to use tools and APIs (p. 38).
- They differ from traditional AI agents by handling complex, multi-step tasks, maintaining state, and adapting to new situations (p. 42).

4. Agentic Design Patterns (pp. 45-58)

- Several design patterns underpin agentic AI:
 - **Planning:** Decomposing complex tasks into manageable subtasks (p. 46).
 - **Tool Use:** Leveraging external resources to extend capabilities (p. 47).
 - **Reflection:** Iteratively evaluating and improving outputs (p. 48).
 - **Agentic RAG:** Integrating agents into RAG workflows for autonomous planning and refinement (p. 49).
 - **Architectures:** Single-agent, multi-agent, parallel, sequential, loop, router, hierarchical, and human-in-the-loop patterns (pp. 51-58).

5. Bootcamp and Learning Path (pp. 39-40)

- The bootcamp covers foundational topics such as transformer architectures, context engineering, vector databases, agentic workflows, and evaluation methods.
 - Emphasis is placed on hands-on learning and building multi-agent LLM applications.
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This summary captures the main themes and structure of "Introduction to Agentic AI" (© 2026 Data Science Dojo). For detailed explanations and examples, refer to the corresponding pages in the original document.